Karan Chawla Dora

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Education

University of Toronto — Toronto, ON
*Bachelor of Applied Science — Engineering Science (cGPA: **3.95/4.0**)* — *2025 – 2030*
Research Interests: *control systems, robot autonomy, state estimation*

Experience

SAE AutoDrive - Toronto Autonomous Vehicle Team (aUToronto) — Sep. 2025 – Present
State Estimation Developer — *Toronto, ON*

- Developed a lane localization pipeline to preserve autonomous vehicle pose accuracy during GPS dropouts.
- Implemented an Iterated Extended Kalman Filter IEKF processing 100+ lane observations per 5 meters.
- Designed ROS2 bag analysis and validation workflows to evaluate performance across simulation and vehicle tests.

FIRST Robotics Competition — Jul. 2021 – Jun. 2025
Robotics Team Captain & Software Development Lead — *Ottawa, ON*

- Directed a 90+ member cross-functional team through design, manufacturing, and software cycles, securing World Championship qualification and 2x Provincial Finalist finishes.
- Developed robust autonomous routines in C++ by integrating Computer Vision (AprilTags), wheel odometry, and trajectory generation paths, increasing autonomous point scoring by 200%.
- Improved robot precision by tuning feedforward and PID feedback control systems for mechanism control.

Research & Publications

Researcher @ Autonomous Space Robotics Lab (ASRL) | *Supervisor: **Tim Barfoot*** — May 2026 – Present

- Implementing a mesh radio network to enable long-range communication for multi-robot fleets.

Bruce Lab — University of Ottawa | *Research Assistant* — Aug. 2024

- Reduced transient time by 80% and steady-state error by 40% by implementing C++ PID controllers for platinum-nickel electrodeposition current and voltage stabilization.
- Accelerated experimental cycles by 33% by architecting a Python GUI/interface for automated trial execution, sensor logging, and data curation.
- Expanded operating voltage range by 5x by refining an Arduino-based potentiostat circuit and control workflow.

Canadian Robotics and Artificial Intelligence Ethics Design Lab | *Robotics Research* — Aug. 2024

- Accelerated project timeline by 50% by characterizing DJI Robomaster S1 behavior in a simulated robotics environment.
- Achieved 1-second target tracking by implementing Python computer vision and machine learning models.

Projects

Reinforcement Learning Q-Learning Trainer Simulator | *Python, FastAPI, NumPy, JavaScript* — June 2026

- Developed a full-stack tabular Q-learning app with custom maps, real-time WebSocket training viz, hyperparameter lab, and model validation suite.
- Deployed RL training dashboard to Fly.io with session isolation and Docker; added pytest regressions for layout parsing, reachability, and evaluation checks.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, TypeScript, HTML, CSS
Robotics & Control: ROS, state estimation, sensor fusion, PID control, odometry, computer vision
Tools & Libraries: Git, Arduino, NumPy, pandas, Matplotlib, Next.js

Awards

HackTheNorth 2024 2x Award Winner: Best Use of MongoDB, Symphonic Labs
2x FRC Provincial Finalist: FIRST Robotics 2023, 2024
Harvard-MIT Math Tournament: Qualified Participant
FIRST Alumni Entrance Scholarship: Value of $15,000
Dean's List Semi-Finalist: FIRST Robotics Competition
University of Toronto Excellence Award (UTEA): Summer research award valued at $7,000